SOARING CLUB

FLIGHT LINE APP

Technical Specification — Version 9

React Native (iOS & Android) • React Web Admin Dashboard • Firebase Firestore • Wild Apricot Integration

1. Overview

The Soaring Club Flight Line App is a two-platform digital system: a React Native mobile app for on-field operations and a React web dashboard for administrative management. Both platforms share a single Firebase Firestore backend and Wild Apricot member sync.

Version 9 confirms: up to four simultaneous tow planes tracked independently via per-session tow pilot records; takeoff time entered by the line chief via a wheels-up tap; landing time entered by either the line chief or the pilot (first tap wins); flight time logging applied uniformly to all flights (club and private) for timing purposes, with billing output only for club gliders; and the admin interface confirmed as a web dashboard separate from the mobile app.

Platform	Purpose
React Native mobile app	On-field use by glider pilots, tow pilots, line chiefs, bookkeepers, and ride givers
React web dashboard	Off-field admin: member management, Wild Apricot sync, aircraft roster, attention queue, flight record correction, reconciliation settings

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2. User Roles

2.1 Glider Pilot (Mobile App)

The glider pilot submits and cancels tow requests. Their two authorities are Submit and Cancel. No certification or clearance step involves the pilot.

Tow Request Form

- Student flight toggle → Solo or Dual (see Section 4 for full form logic)
- Glider selection: club roster or private glider from profile (shows N-number and contest number)
- Requested tow altitude (ft AGL) and tow type; free-text note if Other. Tow type informs the tow pilot what to expect on the tow. Pattern tow is the only type that changes billing and tow pilot workflow.
- Submit — creates the flight record; visible immediately to all connected devices
- Cancel — available any time before takeoff; archives with optional reason

Landing Tap

After their glider touches down, the pilot may tap ‘Log Landing’ on their My Ticket screen. If the line chief has already logged the landing, the pilot sees ‘Landing recorded’ and the tap is suppressed. If neither has tapped, the pilot’s tap writes the landing time. First tap wins.

Club Glider Flight Time Entry

Required for all club glider flights. Not required for private gliders, but the system-calculated flight time (from takeoff and landing taps) is recorded for all flights equally. The pilot enters their declared flight time in decimal hours (tenths). Reminders escalate if not entered (see Section 5).

2.2 Tow Pilot (Mobile App)

The tow pilot's flight-line workflow is passive during flight and unaffected by line-chief mode. The tow pilot additionally owns the global mode toggle and the end-of-day clearance.

Tow Plane Selection at Role Login

Immediately upon selecting the Tow Pilot role, a mandatory tow plane selection screen appears before any other content. Up to four tow pilots may each have an independent active session simultaneously, each with their own tow plane.

- Shows all towPlanes where isActive: true (make, model, N-number)
- Selection creates a towPilotSessions document; the N-number is displayed persistently in the screen header
- The selected tow plane pre-fills the line chief's per-flight tow plane dropdown but is overridable
- To change aircraft: log out of Tow Pilot role and log back in; the warning 'Complete any pending release altitude entries first' is shown
- Up to four concurrent tow pilot sessions are supported; each session is independent

Line-Chief Mode / No-Line-Chief Mode Toggle

A prominent toggle in the tow pilot's header switches the global operating mode stored in globalSettings.lineChiefMode. All connected devices update in real time.

- Line-chief mode ON (default): line chief queue active; takeoff and landing taps record server timestamps; systemFlightTime calculated for all flights
- Line-chief mode OFF: line chief screen read-only with banner; no takeoff/landing timestamps; pilot-entered time is the only source for club gliders; private glider systemFlightTime not calculated
- Mode change banner appears on all devices instantly
- Tow pilot workflow identical in both modes

Release Altitude Entry or Pattern Tow Confirmation

After returning to the ground, the tow pilot opens the completed tow card. The card adapts based on tow type:

- All tow types except pattern: the release altitude field (feet AGL) is shown. Entry and submit is the only data entry the tow pilot makes. With multiple tow planes active, the session links each entry to the correct tow plane automatically.
- Pattern tow: the release altitude field is replaced by a single large button: 'Confirm Pattern Tow.' One tap writes patternTowConfirmed: true. No altitude is entered.

2.3 Line Chief (Mobile App)

Active in line-chief mode. Manages the queue, certifies flights (assigns tow plane, clears for takeoff), logs takeoff, and logs landing. In no-line-chief mode the screen is read-only.

Queue Management

- Pending queue sorted by queuePosition; drag-to-reorder supported
- Can postpone (moves to back of queue) or cancel any pending ticket
- Airborne list shows all flights currently up, using shorthand identifiers (see Section 6)

Flight Certification

For each pending flight before takeoff:

- Line chief selects tow plane from dropdown of active tow planes (pre-filled from tow pilot session; overridable). With up to four active tow planes, all four appear in the dropdown labelled by N-number, make, and model
- Taps ‘Certify — Cleared for Tow’; writes towPlaneId, towPlaneNNumber, certifiedByLineChiefId, certifiedAt; status → certified
- Log Takeoff button activates

Takeoff Time — Wheels-Up Tap

At the moment the glider’s wheels leave the ground, the line chief taps ‘Log Takeoff’ on the certified flight card. This writes takeoffTime as a Firebase serverTimestamp(). The tap is a single large button on the flight card, designed for fast one-handed operation. A confirmation dialog is not shown (speed is essential) but the action can be corrected by Admin within a 5-minute window if mis-tapped.

Landing Time — Tap by Line Chief or Pilot

When the glider touches down, landing time is recorded by whichever of the following happens first:

- Line chief taps ‘Log Landing’ on the airborne flight card — the most common path
- Pilot taps ‘Log Landing’ on their My Ticket screen — available when the line chief is occupied with another tow

The first tap wins. Firestore security rules use a transaction to ensure only one write succeeds; the second tap sees the field already populated and is silently rejected. landingTime is written as a Firebase serverTimestamp(). systemFlightTime is calculated server-side by Cloud Function: (landingTime − takeoffTime) / 3600000, rounded to one decimal place. This applies to all flights — club and private.

Reconciliation

When a pilot submits their pilotFlightTime for a club glider flight, the reconciliation check runs. Discrepancies above tolerance appear on the line chief’s Reconciliation Panel for on-field resolution (see Section 5).

2.4 Bookkeeper (Mobile App)

Read-only. Views billedFlightTime (club gliders) and systemFlightTime (all flights, for reference). Demo rides in a separate tab with ride giver name and member number. Student history accessible. WA account balance reference. CSV/PDF export.

2.5 Admin (Web Dashboard)

The admin interface is a React web application hosted separately from the mobile app (e.g. admin.yourclub.com or a Firebase Hosting URL). It connects to the same Firestore database using the same Firebase project. Admin users authenticate with Firebase Auth using their club email; the admin role flag is checked post-login.

Why a Web Dashboard

Admin tasks — managing member flags, reviewing sync logs, correcting flight records, adjusting reconciliation tolerances, managing the aircraft roster — benefit from a full desktop browser screen. The admin is never required to be on the flight line, so mobile access is not needed for this role.

Admin Dashboard Sections

- Members — search and filter; set/clear isOnNoTowList, isTowPilotQualified, isTowPilotCurrent, isCFIQualified, isCFICurrent, isRideGiverAuthorized; enter/edit cfiNumber and privateGlider.contestNumber
- Wild Apricot Sync — view most recent sync log; trigger manual sync; view error details; configure sync schedule
- Aircraft Roster — add/edit/deactivate club gliders (with isTwoSeat flag) and tow planes (with isActive flag); up to four tow planes may be active simultaneously
- Attention Queue — flights flagged missingFlightTime or needsReconciliation; admin can enter flight time on pilot's behalf or override with reason; all edits audit-logged
- Flight Records — date-range search; view any flight in full detail; correct any field with required editReason; audit trail on all admin writes
- Daily Sessions — view end-of-day close records; reopen a closed day if needed (with reason)
- Settings — adjust reconciliationToleranceMin; configure reminder email timing; manage demo ride account credentials

Admin Tech Stack

The web dashboard is a React application (not React Native) using the Firebase JS SDK (same SDK version as the mobile app). It can be hosted on Firebase Hosting at no additional cost. Authentication is shared with the mobile app — an admin user's Firebase Auth account works on both platforms. The admin web app has its own repository but imports shared Firestore collection constants and type definitions from a shared package.

2.6 Ride Giver / Demo Ride Account (Mobile App)

Accessed via shared Demo Ride account credentials. Routes directly to demo ride form. Ride giver selected from isRideGiverAuthorized members. Ride giver name and member number recorded on every demo flight. Waiver gate mandatory. Two-seat club gliders only.

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3. Multiple Simultaneous Tow Planes (Up to Four)

3.1 Session Architecture

Each active tow pilot has an independent session document in the towPilotSessions collection. Up to four sessions may be open simultaneously, each referencing a different tow plane. The globalSettings document maintains an array of active session references for fast lookup.

Scenario	Handling
1 tow plane	Standard; line chief dropdown pre-fills from the single active session
2–4 tow planes	Line chief dropdown shows all active tow planes by N-number; pre-fills to the session whose tow pilot most recently submitted a brief for this glider; LC can select any active plane
Tow pilot changes plane	Logs out and back in; new session created; prior session closed; prior flights retain original session's tow plane
Tow pilot ends day	Logs out; session closed; remaining active sessions unaffected

3.2 Airborne List with Multiple Tow Planes

The line chief's airborne list shows the tow plane N-number alongside each airborne flight so the line chief can track which aircraft is on which tow simultaneously:

Tow plane	Glider (shorthand)	Status
N2211T	N731SG / Martinez	Airborne 14:32 ↑
N8801K	42 / Smith	Airborne 14:38 ↑
N2211T	N44SP / Okonkwo	Airborne 14:51 ↑

> The tow plane N-number in the airborne list is the value written at certification time by the line chief. If a tow pilot session changes mid-day, the airborne record still shows the plane that actually made that tow.

3.3 globalSettings Active Sessions Array

The globalSettings document maintains activeTowSessions as an array of up to four objects, updated in real time as tow pilots log in and out:

Field	Type	Description
activeTowSessions	array (max 4)	[{ sessionId, towPilotUID, towPilotName, towPlaneId, towPlaneNNumber, sessionStart }]

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4. Tow Request Form Logic

4.1 Decision Tree

studentFlight	isDual	Glider	Outcome
true	false	Any	Student solo → soloHistory on completion
true	true	Any	Dual instruction → CFI selector → instructionHistory
false	false	1-seat	Licensed pilot solo
false	false	2-seat	Licensed pilot solo; back seat empty
false	true	2-seat	Licensed pilot with guest; passenger name required; no waiver

> Dual has two meanings: with studentFlight true, Dual = CFI instruction. With studentFlight false in a two-seat club glider, Dual = guest passenger. Separate code paths; separate fields populated.

4.2 Waiver Rule

The waiver checkbox appears only in the Demo Ride form. No waiver gate for any regular member flight including guest passenger flights.

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5. Flight Timing, Billing, and Reconciliation

5.1 Unified Timing for All Flights

Takeoff and landing taps are applied uniformly to every flight — club glider and private glider alike. This simplifies the codebase: there is no branching on glider type in the timing logic. The billing output stage is where club and private gliders diverge.

Timing field	Club glider	Private glider
takeoffTime	Recorded (line chief wheels-up tap)	Recorded (same)
landingTime	Recorded (LC or pilot tap, first wins)	Recorded (same)

Timing field	Club glider	Private glider
systemFlightTime	Calculated (one decimal hour (tenth))	Calculated (same)
pilotFlightTime	Required entry by pilot	Not required; field left null
billedFlightTime	Set after reconciliation	Always null; not billed
Safety tracking	Cleared on landing + time entry	Cleared on landing tap alone

5.2 Takeoff — Wheels-Up Tap (Line Chief)

The line chief's flight card shows a large, prominent 'WHEELS UP' button once a flight is certified. The line chief taps it at the moment the glider leaves the ground. Firebase serverTimestamp() is written to takeoffTime. No confirmation dialog — speed matters on a busy flight line. Admin has a 5-minute correction window via the web dashboard if mis-tapped.

5.3 Landing — Two-Tap Confirmation Protocol

Landing time is confirmed through a deliberate two-tap protocol involving both the line chief and the pilot. This replaces the first-wins single-tap model with a collaborative confirmation that catches timing discrepancies on the field before the record is committed.

Line-Chief Mode — Two-Tap Flow

Status transitions: airborne → landing_proposed → landed

Step	Detail
Tap 1	Either the line chief (on the airborne flight card) or the pilot (on My Ticket) taps 'Log Landing.' This writes proposedLandingTime and proposedBy to the flight record and sets status: landing_proposed. No landingTime is committed yet.
Notification	The other party receives a real-time banner: 'Landing proposed at [time] by [LC / Pilot] — Confirm or Dispute?' The banner is prominent and persistent until resolved.
Tap 2: Confirm	The second party taps Confirm. landingTime = proposedLandingTime is committed. status: landed. systemFlightTime calculated. landingConfirmMethod: confirmed.
Tap 2: Dispute	The second party taps Dispute and enters their own observed landing time. Both times (proposed and disputed) are displayed side-by-side on the line chief's screen. The line chief selects the accepted time or enters a third corrected value. landingTime is committed with landingConfirmMethod: dispute_resolved. Both proposed times are equal weight — neither is the automatic default.
Timeout	If the second party does not tap within the configurable timeout window (default 5 minutes, stored in globalSettings.landingConfirmTimeoutMin), a Cloud Function auto-confirms the proposed time. landingTime = proposedLandingTime; landingConfirmMethod: auto_timeout. This prevents stalled records from blocking end-of-day close.

> In a dispute, the line chief has the final say on which time is accepted — they are physically watching the landing pattern. However, both observed times are equal weight inputs; neither is pre-selected as the default. The line chief may also enter a third corrected value if they judge both entries to be inaccurate.

No-Line-Chief Mode — Single Tap

With no line chief present, the pilot is the sole observer of their own landing. The two-tap protocol does not apply. The pilot taps ‘Log Landing’ once on their My Ticket screen and landingTime is committed immediately. landingConfirmMethod: no_lc_single_tap. No systemFlightTime is calculated (no takeoff timestamp exists in no-LC mode). The pilot then enters their declared flight time as the only time record for club gliders.

Common to All Paths

- Both landing buttons (line chief and pilot) are large, prominent, and designed for fast one-handed outdoor use
- Once landingTime is committed by any method, it is locked. Admin can correct via web dashboard with a required editReason and audit entry
- systemFlightTime is calculated by Cloud Function immediately after landingTime is committed, for all flights in line-chief mode

5.4 Pilot Flight Time Entry (Club Gliders)

After landing is recorded, the pilot’s My Ticket screen for a club glider flight shows a required ‘Your flight time’ field (decimal hours tenths, e.g. 0.5 for 30 minutes, 1.5 for 90 minutes). This is the primary billing input.

Reminder Escalation

Trigger	Channel	Action
Landing recorded; pilotFlightTime null	In-app banner	Persistent on My Ticket: ‘Enter your flight time to complete your record’
2 hours after landing; still null	Email (Cloud Function)	Automated email with deep link to flight entry screen
24 hours after landing; still null	Admin flag	missingFlightTime: true; appears in web dashboard Attention Queue

5.5 Reconciliation Workflow

Applies to club glider flights in line-chief mode only. Private gliders are never reconciled (billedFlightTime is always null).

Step	Detail
1	Pilot submits pilotFlightTime. Cloud Function reads systemFlightTime.

Step	Detail
2	Compute diff = |pilotFlightTime − systemFlightTime|
3a	diff ≤ reconciliationToleranceMin (default 2.0): billedFlightTime = pilotFlightTime; reconciled: true; method: auto
3b	diff > tolerance: needsReconciliation: true; both values shown on line chief Reconciliation Panel and Admin Attention Queue
4	Line chief reviews on-field: accepts pilot time, accepts system time, or enters corrected value. resolvedByUID and resolvedAt written. billedFlightTime locked.
5	reconciled: true written. Only Admin (web dashboard) can subsequently change billedFlightTime, with required editReason and audit entry.

> In no-line-chief mode: systemFlightTime is not calculated. billedFlightTime = pilotFlightTime on submission. reconciled: true; method: no_lc_self_log. No comparison step.

5.6 Pattern Tow — Special Rules

A pattern tow (towType: pattern) is a brief tow to pattern altitude, typically used for box-the-wake or slack line recovery practice. It has distinct rules for the tow pilot and for billing.

Tow Pilot: Confirm Pattern Tow (No Altitude Entry)

When the towType on a flight record is pattern, the tow pilot's completed tow card shows a single large button: 'Confirm Pattern Tow' instead of the release altitude entry field. There is no altitude to enter — the tow pilot taps the button and is done.

- Tapping Confirm Pattern Tow writes patternTowConfirmed: true and patternTowConfirmedAt (server timestamp) to the flight record
- releaseAltitudeFt remains null for all pattern tows
- The confirmation is required before the tow record can be marked complete; the same escalation logic as pilot flight time applies if not confirmed

TOL Recording

The two-tap landing protocol applies normally to pattern tows. Takeoff and landing times are recorded as usual. The actual duration is captured in systemFlightTime for safety tracking and end-of-day clearance. The billed value, however, is fixed.

Billing Rules by Flight Type

towType	Purpose (informs tow pilot)	Release alt	billedFlightTime	Pilot time entry?
pattern	Short pattern tow; flat rate	None; Confirm tap	Always 0.1 hr	No — auto-set
normal	Standard altitude tow	Entered by TP	Actual TOL time	Yes (club glider)

towType	Purpose (informs tow pilot)	Release alt	billedFlightTime	Pilot time entry?
box_wake	Box the wake on tow	Entered by TP	Actual TOL time	Yes (club glider)
slack_line	Slack line recovery on tow	Entered by TP	Actual TOL time	Yes (club glider)
tow_rope_fail	Tow rope break practice	Entered by TP	Actual TOL time	Yes (club glider)
other	Free-text note to tow pilot	Entered by TP	Actual TOL time	Yes (club glider)

> Pattern tow billing is a flat 0.1 hr for every flight regardless of who is flying — student, instructor, or licensed member. The tow type is informational only for the tow pilot. Box the wake, slack line recovery, and tow rope fail tows go to altitude and bill actual TOL time like any normal tow; the label simply tells the tow pilot what maneuver to expect during the climb.

Reconciliation for Pattern Tows

- All pattern tows: billedFlightTime is set to 0.1 hr by Cloud Function automatically on patternTowConfirmed write; reconciled: true; reconciliationMethod: pattern_tow_fixed. No pilot time entry required. No reconciliation step. No exceptions.

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6. Private Glider Display, Contest Numbers, and Safety Tracking

6.1 Private Glider Member Record

Field	Type	Description
privateGlider.nNumber	string	FAA registration (e.g. N5742K)
privateGlider.make	string	Manufacturer
privateGlider.model	string	Model
privateGlider.contestNumber	string|null	2 or 3 digit competition number (e.g. 42, WB)

6.2 Shorthand Display Rule

Context	Club glider	Private glider
Tow pilot brief (airborne)	N731SG / Martinez	42 / Smith
Line chief airborne list	N731SG / Martinez	42 / Smith
End-of-day safety list	N731SG / Martinez ✓	42 / Smith ✓
Completed flight record	Full detail	Full detail

Fallback if contestNumber is null: N-number + last name. displayShorthand is pre-computed and stored on the flight record at submission time so list rendering never needs to join member data.

6.3 Private Glider Safety Tracking

Because all flights record takeoffTime and landingTime uniformly, private glider flights are automatically cleared once landingTime is written. No pilot flight time entry is required. The end-of-day checklist shows private glider flights as Cleared as soon as landing is logged, making safety tracking seamless with no extra workflow.

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7. End-of-Day Safety Clearance

7.1 Clearance Responsibility

Either the line chief or the tow pilot may close the flying day. Both have access to the End-of-Day Checklist screen. The checklist is also visible (read-only) in the Admin web dashboard.

7.2 Flight Status Categories

Status	Meaning
Cleared ✓	Club glider: landed + pilotFlightTime entered + reconciled. Private glider: landed (landingTime recorded). Pilot accounted for.
Pending ⚠	Still airborne, or club glider missing pilotFlightTime, or unreconciled discrepancy.
Cancelled ✕	Tow cancelled before takeoff. No flight. No further action.

7.3 Resolving Pending Flights

- Still airborne / no landing recorded: line chief or tow pilot physically confirms pilot is on the ground, then taps ‘Confirm pilot returned safely’; writes manualClearanceAt and manualClearedByUID
- Missing pilotFlightTime (club glider): line chief or tow pilot enters time on pilot’s behalf with a note; or waives entry with override and reason (Admin can also do this from the web dashboard)
- Unreconciled discrepancy: resolve via Reconciliation Panel before closing

7.4 Close Flying Day

Once all flights are Cleared or Cancelled, ‘Close Flying Day’ becomes active. The closer taps it, optionally adds a note, and confirms. Writes dayClosedAt, dayClosedByUID, dayClosedMode to the dailySessions document. The day's queue is locked from further modifications (Admin can reopen from the web dashboard if needed).

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8. Data Model (Firestore) — Complete

8.1 globalSettings document (globalSettings/current)

Field	Type	Description
lineChiefMode	boolean	true = line chief mode; toggled by tow pilot
modeChangedAt	timestamp	Server timestamp of last mode change
modeChangedByUID	string	UID of tow pilot who toggled
activeTowSessions	array	Up to 4 objects: { sessionId, towPilotUID, towPilotName, towPlaneId, towPlaneNNumber, sessionStart }
reconciliationToleranceMin	number	Default 2.0 decimal hours (tenths); Admin-adjustable via web dashboard
reminderEmailDelayHours	number	Default 2; hours after landing before reminder email
missingTimeFlagHours	number	Default 24; hours before missingFlightTime flag set
landingConfirmTimeoutMin	number	Default 5; minutes before proposed landing auto-confirms if second party does not tap

8.2 towPilotSessions collection

Field	Type	Description
sessionId	string	Auto-generated
towPilotUID	string	UID of tow pilot
towPilotName	string	Denormalized name
towPlaneId	string	Ref to towPlanes
towPlaneNNumber	string	Denormalized tail number
sessionStart	timestamp	When session began
sessionEnd	timestamp|null	When tow pilot logged out
isActive	boolean	True while session is open

8.3 members collection

Field	Type	Description
memberId	string	Firebase Auth UID
wildApricotId	string	WA contact ID — sync key

Field	Type	Description
memberNumber	string	Club member number
displayName	string	Full name
email	string	Login email
membershipStatus	string	Active|Lapsed|Pending (from WA)
accountBalance	number	WA balance (read-only)
renewalDue	timestamp	From WA
roles	string[]	glider_pilot|tow_pilot|line_chief|bookkeeper|admin|demo_ride_account
isTowPilotQualified	boolean	Admin-set
isTowPilotCurrent	boolean	Admin-set
isCFIQualified	boolean	Admin-set
isCFICurrent	boolean	Admin-set
cfiNumber	string|null	FAA CFI certificate number
isOnNoTowList	boolean	Admin-set: blocks tow submission
isRideGiverAuthorized	boolean	Admin-set
isDemoRideAccount	boolean	True for the shared demo ride login
privateGlider	object|null	{ nNumber, make, model, contestNumber }
lastSyncedAt	timestamp	Last WA sync

8.4 gliders collection

Field	Type	Description
gliderId	string	Auto-generated
nNumber	string	FAA registration
make	string	Manufacturer
model	string	Model
isTwoSeat	boolean	Enables passenger fields and demo ride filter
isActive	boolean	Shown in pilot glider selector

8.5 towPlanes collection

Field	Type	Description
towPlaneId	string	Auto-generated
nNumber	string	FAA registration
make	string	Manufacturer
model	string	Model

Field	Type	Description
isActive	boolean	Shown in tow plane selectors; up to 4 may be active

8.6 flights collection (complete)

Field	Type	Description
flightId	string	Auto-generated
status	string	pending|certified|airborne|landing_proposed|landed|complete|postponed|c
queuePosition	number	LC-managed sort order in pending queue
createdAt	timestamp	Server timestamp: submission
updatedAt	timestamp	Server timestamp: most recent write
pilotId	string	UID of glider pilot
pilotName	string	Denormalized name
memberNumber	string	Denormalized member number
studentFlight	boolean	True if designated a student flight
isDual	boolean	True if second occupant (CFI or guest)
instructorId	string|null	CFI UID (studentFlight + isDual only)
instructorName	string|null	Denormalized CFI name
cfiNumber	string|null	Snapshot CFI FAA number
hasPassenger	boolean	True when non-student isDual (guest)
passenger.name	string|null	Guest name
isDemoRide	boolean	True for demo ride flights
rideGiverId	string|null	UID of ride giver
rideGiverName	string|null	Ride giver name
rideGiverMemberNumber	string|null	Ride giver member number
demoRide	object|null	{ passengerFirstName, passengerLastName, passengerContact, waiverC waiverConfirmedAt, paymentNote }
gliderId	string|null	Ref to gliders; null for private
gliderNNumber	string	Denormalized N-number
gliderOwnership	string	club|private
displayShorthand	string	Pre-computed: ‘42 / Smith’ or ‘N731SG / Martinez’
requestedAltitudeFt	number	Pilot-requested tow altitude AGL
towType	string	pattern|normal|box_wake|slack_line|tow_rope_fail|other
towTypeNote	string|null	Free text if towType === other
towPilotSessionId	string|null	Ref to towPilotSessions; links flight to tow plane
towPilotUID	string|null	UID of tow pilot who enters release alt
towPilotName	string|null	Denormalized

Field	Type	Description
towPlaneId	string|null	Ref to towPlanes; set by LC at certification
towPlaneNNumber	string|null	Denormalized; set by LC at certification
certifiedByLineChiefId	string|null	UID of certifying LC
certifiedAt	timestamp|null	Certification server timestamp
releaseAltitudeFt	number|null	Actual release alt; null for pattern tows
patternTowConfirmed	boolean	True when tow pilot taps Confirm Pattern Tow (pattern tows only)
patternTowConfirmedAt	timestamp|null	Server timestamp of pattern tow confirmation
takeoffTime	timestamp|null	Wheels-up server timestamp; LC tap
proposedLandingTime	timestamp|null	Server timestamp from Tap 1 (first tapper); not yet committed
proposedBy	string|null	line_chief | pilot — who made Tap 1
proposedAt	timestamp|null	Server timestamp when Tap 1 was made
landingDisputedTime	timestamp|null	Second party's observed landing time, entered on Dispute
landingDisputedBy	string|null	line_chief | pilot — who disputed
landingTime	timestamp|null	Committed touchdown time; set only after Confirm, LC dispute resolution,
landingLoggedBy	string|null	line_chief | pilot — who made Tap 1
landingConfirmMethod	string|null	confirmed | dispute_resolved | auto_timeout | no_lc_single_tap
landingConfirmTimeoutMin	number|null	Snapshot of globalSettings timeout at time of Tap 1 (default 5)
systemFlightTime	number|null	(landingTime−takeoffTime)/3600000 one decimal place hours; all flights
pilotFlightTime	number|null	Pilot-entered decimal hours (tenths); club gliders required
pilotFlightTimeAt	timestamp|null	When pilot submitted
billedFlightTime	number|null	Final accepted billing value; null for private gliders
reconciled	boolean	True once billing value is final
reconciliationMethod	string|null	auto|accepted_pilot|accepted_system|manual_correction|no_lc_self_log|p
needsReconciliation	boolean	True when diff > tolerance and unresolved
resolvedByUID	string|null	UID of resolver
resolvedAt	timestamp|null	Resolution timestamp
missingFlightTime	boolean	True after 24hrs; club glider; no pilot entry
reminderEmailSentAt	timestamp|null	2hr reminder email timestamp
lineChiefPresent	boolean	Snapshot: LC active at submission
manualClearanceAt	timestamp|null	Set when LC/TP manually confirms pilot safe
manualClearedByUID	string|null	UID of manual clearer
postponeCount	number	Times postponed
cancelledAt	timestamp|null	Set on cancellation
cancelledBy	string|null	UID of canceller
cancelReason	string|null	Optional reason

8.7 dailySessions collection

Field	Type	Description
sessionDate	string	ISO date YYYY-MM-DD
dayClosedAt	timestamp|null	Close timestamp
dayClosedByUID	string|null	Closer UID
dayClosedMode	string|null	line_chief | tow_pilot
totalFlights	number	All flights for the day
totalCleared	number	Cleared count at close
totalCancelled	number	Cancelled count
notes	string|null	Optional closer note

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9. Complete Per-Tow Workflow

9.1 Line-Chief Mode

Step	Actor	Tap / Action	Data written
1	Glider pilot	Submits tow request	Flight doc created; status: pending
2	Tow pilot	Brief appears (passive)	No write
3	Line chief	Selects tow plane; taps Certify	towPlaneNNumber; certifiedAt; status: certified
4	Line chief	Taps WHEELS UP at wheels-up	takeoffTime (serverTimestamp); status: airborne
5	Tow pilot	Tows — no app action	(none)
6	Tow pilot	On ground: enters release altitude OR taps Confirm Pattern Tow	releaseAltitudeFt (non-pattern) OR patternTowConfirmed: true (pattern)
7a	LC or Pilot	TAP 1: Taps Log Landing — proposes time	proposedLandingTime; proposedBy; status: landing_proposed
7b	Other party	TAP 2: Confirm — accepts proposed time	landingTime committed; method: confirmed; systemFlightTime
7b*	Other party	TAP 2: Dispute — enters own observed time; LC selects final	landingDisputedTime; LC selects; landingTime; method: dispute_resolved

Step	Actor	Tap / Action	Data written
7b**	Cloud Function	Timeout (5 min default) if no Tap 2	landingTime = proposedLandingTime; method: auto_timeout
8	Glider pilot	Enters pilotFlightTime (club only)	pilotFlightTime; triggers reconciliation CF
9	Cloud Function	Reconciliation check	billedFlightTime or needsReconciliation flag
10	Line chief (if needed)	Resolves on Reconciliation Panel	billedFlightTime; reconciled: true; locked

9.2 No-Line-Chief Mode

Step	Actor	Tap / Action	Data written
1	Glider pilot	Submits tow request	Flight doc created; status: pending
2	Tow pilot	Brief appears; tows; enters release alt on ground	releaseAltitudeFt
3	Pilot	Single tap Log Landing — no confirmation needed	landingTime committed; method: no_lc_single_tap
4	Pilot	Enters pilotFlightTime (club only)	billedFlightTime = pilotFlightTime; method: no_lc_self_log

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10. Screen Map

10.1 Mobile App Screens

Screen	Role(s)	Purpose
Login	All	Email/password; demo account routes direct to demo form
Role Select	Multi-role	Choose session role
Tow Plane Select (gate)	Tow Pilot	Mandatory; all active tow planes listed; header set
Tow Brief + Mode Toggle	Tow Pilot	Live briefs; shorthand IDs; Line Chief On/Off switch; release alt entry
End-of-Day Checklist	Tow Pilot, LC	All flights Cleared/Pending/Cancelled; Close Flying Day
Tow Request Form	Glider Pilot	Student/dual/passenger branch; altitude; type; Submit/Cancel

Screen	Role(s)	Purpose
My Ticket	Glider Pilot	Status; queue position; Log Landing tap; flight time entry; cancel
Demo Ride Form	Demo Acct	Ride giver select; passenger; waiver gate; two-seat glider
Flight Queue (LC mode)	Line Chief	Pending (reorder, certify, WHEELS UP) + Airborne (shorthand + tow plane); Reconciliation Panel
Flight Queue (no-LC mode)	Line Chief	Read-only with mode banner
Daily Log — Flights	Bookkeeper	billedFlightTime; systemFlightTime; all fields; date filter
Daily Log — Demo Rides	Bookkeeper	Ride giver name+number; passenger; billing notes
Student History	Bookkeeper	Dual/solo decimal hours; instructors; tow types
Member Profile	All	Name; member#; private glider N-number + contest number

10.2 Web Dashboard Sections (Admin)

Section	Purpose
Members	Search/filter all members; set all flags; enter cfiNumber and contestNumber; view WA sync status per member
Wild Apricot Sync	Most recent sync log; manual trigger; error details; sync schedule config
Aircraft Roster	Add/edit/deactivate gliders (isTwoSeat) and tow planes (isActive); up to 4 tow planes active
Attention Queue	missingFlightTime and needsReconciliation flags; enter time on pilot's behalf; override with reason
Flight Records	Date-range search; full detail view; correct any field with editReason; audit trail
Daily Sessions	View close records; reopen closed day with reason
Settings	reconciliationToleranceMin; reminder email timing; demo account management

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11. Technology Stack

11.1 Mobile App

Layer	Technology
Framework	React Native (Expo) + TypeScript
Navigation	React Navigation v7; role-based routing; tow plane gate screen

Layer	Technology
Database	Firebase Firestore JS SDK; real-time onSnapshot listeners; offline persistence
Auth	Firebase Auth; email/password; role + flag check post-login
Global mode	globalSettings onSnapshot listener on all devices
Landing transaction	Firestore transaction for first-wins landing tap
Drag-to-reorder	react-native-draggable-flatlist; atomic batch write on drag-end
Forms	React Hook Form; decimal hour (tenth) validation; waiver gate
Offline	Firestore enablePersistence(); offline banner
Push	FCM — mode change alerts; tow pilot new-brief alert
CI/CD	Expo EAS Build; OTA updates; TestFlight + Play Store

11.2 Admin Web Dashboard

Layer	Technology
Framework	React (Vite) + TypeScript
Database	Firebase Firestore JS SDK (same project as mobile app)
Auth	Firebase Auth (shared; admin role flag enforced post-login)
Hosting	Firebase Hosting (e.g. admin.yourclub.com)
UI	React + Tailwind CSS; data tables for member and flight lists
Shared code	Shared npm package: Firestore collection names, field constants, TypeScript types

11.3 Backend (Firebase Cloud Functions)

Function	Trigger & Purpose
onLandingRecorded	Firestore write to landingTime: calculate systemFlightTime for all flights
onPilotTimeSubmitted	Firestore write to pilotFlightTime: run reconciliation comparison; set billedFlightTime or needsReconciliation
onPatternTowConfirmed	Firestore write to patternTowConfirmed: set billedFlightTime: 0.1 hr and reconciled: true; reconciliationMethod: pattern_tow_fixed. No flag evaluation needed — flat rate applies to all pattern tows without exception.
onFlightComplete	Flight marked complete: write instructionHistory or soloHistory subcollection
reminderEmailScheduler	Runs every 30min: find club glider flights with null pilotFlightTime past 2hr threshold; send email via SendGrid
missingTimeFlagScheduler	Runs every hour: find club glider flights null pilotFlightTime past 24hrs; set missingFlightTime: true

Function	Trigger & Purpose
wildApricotSync	Cloud Scheduler (every 20min during ops hours): fetch WA members; upsert Firestore; write syncLog

12. Development Phases

Phase 1 — Core Mobile MVP (8–10 weeks)

- Firebase project; Firestore schema; security rules; globalSettings listener
- Tow pilot: tow plane gate (all active planes listed); mode toggle; passive brief; release alt on ground; multi-session support for up to 4 planes
- Glider pilot: tow request form (student/dual/passenger branch); Submit and Cancel; Log Landing tap; club glider flight time entry with in-app reminder
- Line chief: pending queue (drag-to-reorder); certification (tow plane dropdown showing all active); WHEELS UP tap; airborne list with tow plane + shorthand; Log Landing; reconciliation panel
- Firestore landing transaction (first-wins)
- End-of-day checklist and Close Flying Day
- Demo ride form; ride giver selector; waiver gate
- Shorthand display (contest# / last name) pre-computed at submission

Phase 2 — Cloud Functions + Reconciliation (3–4 weeks)

- onLandingRecorded: systemFlightTime for all flights
- onPilotTimeSubmitted: reconciliation check; auto or flag
- Reminder email scheduler (SendGrid)
- missingFlightTime flag scheduler
- onFlightComplete: instructionHistory / soloHistory

Phase 3 — Admin Web Dashboard (4–5 weeks)

- React + Vite web app; Firebase Hosting deployment
- Members section: all flags; CFI number; contest number
- Attention Queue: missing time + unreconciled flags
- Aircraft Roster: gliders and tow planes management
- Flight Records: date-range search; correction with audit trail
- Settings: tolerance; reminder timing

Phase 4 — Wild Apricot Integration (3–4 weeks)

- WA sync Cloud Function; app-only flags protected
- WA Sync section in admin dashboard: log viewer; manual trigger
- Membership gate at login; no-tow list enforcement

Phase 5 — Polish & Export (2–3 weeks)

- CSV/PDF export (bookkeeper); student history screen
- FCM push to tow pilot; mode change notification
- Offline hardening; Admin correction 5-minute window for mis-tapped takeoff

Phase 6 — Enhancements

- WA billing write-back; instructor endorsement PDF; solo auth workflow; METAR; winch launch

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13. Notes

The shared Firestore database means the admin web dashboard and the mobile app are always looking at the same live data. An admin correcting a flight record on their laptop is immediately reflected on the line chief's phone without any sync delay. This is a key advantage of the Firebase architecture.

The two-tap TOL protocol replaces the previous first-wins transaction model with a deliberate confirmation workflow. The proposedLandingTime is written on Tap 1 as a non-committed value; landingTime is only committed after Confirm, dispute resolution, or auto-timeout. This means there is no race condition to prevent — Tap 1 always writes (proposedLandingTime) and Tap 2 always resolves (landingTime). The Cloud Function auto-timeout is scheduled using Firebase Cloud Tasks, which creates a task at Tap 1 time that fires after globalSettings.landingConfirmTimeoutMin if landingTime is still null.

Up to four active tow plane sessions is a design limit, not a technical limit. The activeTowSessions array in globalSettings could hold more. Four is chosen as a practical maximum for a typical soaring club; if the club ever runs more than four tow planes simultaneously, this constant can be increased by Admin without a code change.

Contest numbers are not managed by Wild Apricot and must never be overwritten by the WA sync. The sync Cloud Function explicitly skips the privateGlider field on member documents. Pilots or Admin enter contest numbers via the Member Profile screen or the admin web dashboard respectively.

This document should be reviewed with the club's tow pilots, line chiefs, operations manager, bookkeeper, and a developer before work begins.

Version 9 changes: (1) all flight time units converted to decimal hours tenths (0.1 hr = 6 min, 0.5 hr = 30 min, 1.5 hr = 90 min); (2) pattern tow tow-pilot workflow uses Confirm Pattern Tow button instead of release altitude entry; (3) pattern tow billing fixed at 0.1 hr except for dual instruction pattern tows which record actual time; (4) two-tap TOL protocol from v7 retained throughout.